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NPS/002/015 – Technical Specification for 11kV and 20kV Cable Joints & Terminations

1. Purpose

The purpose of this document is to detail the technical requirements of 11 & 20kV cable joints and terminations for use on CE Electric UK's distribution network.
This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/015	1	Specification for 11 and 20kV Cable Joints and Terminations

2. Scope

This specification details the technical requirements for 11kV and 20kV cable joints and terminations for use on CE Electric UK's distribution network. Accessories specified in this document shall be suitable for use on cables installed in accordance with the CE Electric UK policy for the installation of cables: DSS/026.

3. Technical Requirements

3.1.1 Conditions of Operation for Cable Joints.

The following are the general conditions under which cable joints purchased in accordance with this specification are required to operate:

- Nominal system voltages: 11kV and 20kV.
- The working voltage of any part of the system does not normally exceed the normal system voltage by more than 10%.
- Nominal system frequency: 50Hz
- The system operates with the neutral point earthed either directly or through a resistance or reactance at one or more points.

3.1.2 Joints

Cable joints are required to make a physical and electrical connection between cables of various sizes and types. Existing cables types are :

- Single or three-core
- Paper or XLPE insulated
- Have aluminium (solid and stranded) or stranded copper conductors
- May contain water swellable tapes or powders
- Have a lead or aluminium sheaths with or without armour serving, or no metallic sheath
- Be PVC or PE over sheathed.

Single core joints shall be cold applied systems. Joints containing three core paper cables shall be encapsulated in a filling medium such as resin or equivalent alternative.

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3.1.3 Terminations and Insulation Boots

Terminations will be required for indoor and outdoor use. Indoor terminations will normally be fitted into air insulated cable boxes and insulated shrouding boots may be required where sufficient clearances cannot be obtained.

3.1.4 Connectors, Lugs and Earthing Connections

Connectors and lugs shall be blocked if joining paper cables. Connectors will be tested to ENATS C79 or equivalent. Lugs and connectors will be suitable for copper or aluminium conductors. Lugs for outdoor use will be two holes and suitable for an outdoor environment. Earthing connections shall be tested to ENATS C93 or equivalent

3.1.5 Joint Shells

Joint shells shall be plastic, a minimum of 3mm thick with ribbing for added strength and to assist the release of any trapped air. The joint shells are required to have stepped ports and must be provided with any required fasteners and adequate means of sealing. The supplier shall advise the resin volumes for each joint shell.

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3.2 20kV Cable joint and Terminations

All new installations currently utilise single core polymeric cables to BS 7870-4, the range of which is detailed in Appendix 1.

3.2.1 Terminations

Indoor and outdoor terminations, which shall be supplied in kit form, are to be suitable for polymeric cables. Outdoor terminations will be used where they will be exposed to solar radiation and weathering. The terminations shall be cold applied and contain all the necessary items including mechanical lugs to carry out the connection. The terminations shall be tested to BS 7888 and Cenelec HD629 or equivalent.

3.2.2 Insulation Shrouding Boots.

Additional shrouding boots will be cold applied and such that they will fit both straight and angled switch gear bushings. Shrouding boots shall be tested to BS 7888 and HD629.

3.2.3 Insulated Pre-moulded Separable Connectors

Separable connectors will be tested according to BS 7888 and HD 629, and required to fit Cenelec EN 50180 and EN 50181 interfaces.

Separable connectors will be 250-amp push fit non-load break/make and 630 amp bolted type.

3.2.4 Earth Connections

Earth connections will be required to carry 2kA for 3 seconds and should comply with ENATS C93 or equivalent.

3.2.5 Joints

Joints shall be tested to BS 7888 and Cenelec HD629 or equivalent and will be required to connect cable combinations as detailed in section 3.1.2. The joint, which shall be in kit form, shall include all the necessary items including mechanical connectors to carry out the connection. Cold applied joints shall be provided for polymeric to polymeric cables.

3.2.6 Stop Ends

A stop end kit as detailed in BS 7888 is required for existing paper cable to BS 7894 and polymeric cables to BS 7870-4, cable range and size are as detailed in Appendix 1. Cold applied stop ends shall be provided for polymeric cables.

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3.3 11kV Cable Joints and Terminations

All new installations currently utilise single core polymeric cables to BS 7870-4, the range of which is detailed in Appendix 1. All 11kV joints and stop ends will be tested to BS 7888 and Cenelec HD 629 or equivalent.

3.3.1 Terminations

Indoor and outdoor terminations, which shall be supplied in kit form, are to be suitable for polymeric cables. Outdoor terminations will be used where they will be exposed to solar radiation and weathering. The terminations shall be cold applied and contain all the necessary items including mechanical lugs to carry out the connection. The terminations shall be tested to BS 7888 and Cenelec HD629 or equivalent.

3.3.2 Insulation Shrouding Boots.

Additional shrouding boots will be cold applied and such that they will fit both straight and angled switchgear bushings. Shrouding boots shall be tested to BS 7888 and HD 629.

3.3.3 Insulated Pre-moulded Separable Connectors

Separable connectors will be tested according to BS 7888 and HD 629, and required to fit Cenelec EN 50180 and EN 50181 interfaces.

Separable connectors will be 250-amp push fit non-load break/make and 630 amp bolted type.

3.3.4 Earth Connections

Earth connections will be required to carry 5kA for 2 seconds and should comply with EA C93 or equivalent.

3.3.5 Joints

Joints will be required to connect single core polymeric to single core polymeric in straight and branch joint configurations.

Joints will be required to connect single core polymeric onto three core paper insulated cables, which may have an aluminium or lead sheath.

Joints will be required to connect three core paper cables to three core paper cables, which may have aluminium or lead sheath.

3.3.6 Stop Ends

A stop end kit as detailed in BS 7888 is required for existing paper cable to BS 7894 and polymeric cables to BS 7870-4, cable range and size are as detailed in Appendix 1. Cold applied stop ends shall be provided for polymeric cables.

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4 References

4.1 External Documentation

The joints and terminations shall comply with all current versions of the relevant International Standards, British Standard Specifications and all relevant Energy Network Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant

Reference	Title	
Energy Network Association Technical Specification		
ENATS-C79	Type Approvals tests for Connectors and Terminations for Aluminium conductors of insulated power cables.	
ENATS-C93	Type Approval tests for mechanical connections to Metallic sheaths of cables.	
ENATS 09-12	Impregnated Paper Insulated Corrugated Aluminium Sheathed 6350 / 11000v Cable.	
British Standards		
BS-7888	Test requirements on power cables accessories with rated voltage from 3.6/6kV(Um= 7.2 kV up to and including 20.8/36 kV (Um = 42kV).	
BS-6480	Specification for impregnated paper-insulated lead or lead alloy sheathed electric cables of rated voltage up to and including 33000 V.	
BS-7870-4	LV and MV polymeric cables. Specification for distribution cables with extruded insulation for rated voltages of 11 and 33kV.	
BS-7894	MV Impregnated Paper Insulated Cable 3.8/6.6kV to 19/33kV.	
BS EN-50180	Bushings above 1 kV up to 36 kV and from 250 A to 3,15 kA for liquid filled transformers.	
BS EN-50181	Plug-in type bushings above 1 kV up to 36 kV and from 250 A to 1.25 kA for equipment other than liquid filled transformers.	
IEC Technical Specification		
Cenelec HD629	Test requirements on accessories for use on power cables with rated voltages from 3.6/6kV(Um = 7.2 kV) up to and including 20.8/36 (Um = 42kV.	
	Part 1	Cables with extruded insulation.
	Part 2	Cables with unpregnated paper insulation.

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

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4.2 Internal Documentation

Reference	Title
NIL	

5. Definitions

Term	Definition
NIL	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	27/10/08

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	System Strategy	M Nicholson	30/10/08

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	27/10/08

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	21/11/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	10/11/08

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	10/11/08

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Appendix 1 – Typical cable types and sizes (or imperial equivalents) to be found on the CE Electric UK's distribution network

20kV

95 mm² Stranded Aluminium Conductor Cable, Screened, Earthed, PILC SWA, Served, 3-Core

185mm²: Stranded Aluminium Conductor Cable, Screened, Earthed, PILC SWA, Served, 3-Core

95 mm²: Single Core XPLE: Solid Aluminium core and copper wire screens

150 mm²: Single Core XLPE: Stranded Aluminium core and copper wire screens

185 mm²: Single Core XLPE: Solid Aluminium core and copper wire screens

11kV

95 mm²: Aluminium Conductor, earthed, Belted, PI Corrugated Aluminium Sheath, and PVC over sheath.

185 mm²: Aluminium Conductor, earthed, Belted, PI Corrugated Aluminium Sheath, and PVC over sheath

300 mm²: Aluminium Conductor, earthed, Belted, PI Corrugated Aluminium Sheath, and PVC over sheath

95 mm²: Single Core XPLE: Solid Aluminium core and copper wire screens

150 mm²: Single Core XLPE: Stranded Aluminium core and copper wire screens

185 mm²: Single Core XLPE: Solid Aluminium core and copper wire screens

240 mm²: Single Core XLPE: Stranded Aluminium core and copper wire screens

300 mm²: Single Core XLPE: Solid Aluminium core and copper wire screens

400 mm²: Single Core XLPE: Circular Stranded Copper core and copper wire screens

630 mm²: Single Core XLPE: Circular Stranded Copper core and copper wire screens

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Appendix 2 – Addendum to Supplier Requirements

Work Instructions

Kits supplied shall be provided with work instructions in an agreed format.

Kitting

All joint and terminations will be supplied in kit form with the components required including connectors, lugs and work instructions included in the kit.

Tooling

The tooling costs associated with the handling, installation, inspection, maintenance, repair and decommissioning of the product(s) is crucial to CE Electric UK's assessment of the product viability. To enable CE Electric UK to understand and assess the cost of required tooling the Tenderer should provide full details of the tooling recommended for use by the manufacturer for the purpose of handling, installation, commissioning, inspection, maintenance, repair and decommissioning of each tendered product. Where available the Tenderer should provide indicative prices applicable to the recommended tooling.

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Appendix 3 – Logistical Requirements

Storage

To enable CE Electric UK to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment, maximum stacking height with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to:

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

Packaging and labelling

The Tenderer shall ensure that each item is suitably packaged and protected to maintain the product and packaging as "fit for service" prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the Tenderer is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where products packages tendered are made up of sub packages each sub packages shall be marked. As a minimum requirement the following shall be included:

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * CE Electric UK product code
- * Weight

Tenderer shall submit at the time of tendering a sample of the proposed labelling for each product package type.

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Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

Specification for 11kV and 20kV Cable Joints and Terminations required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the Joint / Termination Type (Voltage, Cable Type and Size)

Name:

Signature:

Date:

NOTE:.

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Specification for 11kV and 20kV Cable Joints and Terminations - I/D Paper Terminations				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Connectors / Lugs	C79		
	Sheath (Earth) Connectors	C93		
	I/D Paper Terminations	(HD 629.2)		
	- DC Withstand	Table 2 (1) (15min x 6U ₀)		
	- AC Withstand	Table 2 (2) (5min x 4.5U ₀)		
	- Impulse Withstand (@ elevated temp)	Table 2 (3) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))		
	- Heat Cycle at Voltage	Table 2 (4) (126 cycles x 1.5U ₀)		
	- AC Withstand	Table 2 (5) (4hrs x 3U ₀)		
	- Thermal S/C-screen	Table 2 (6) (2 x I _{sc})		
	- Thermal S/C-conductor	Table 2 (7) (2 x Θ_{sc})		
	- Dynamic S/C (Class 1)	Table 2 (8) (1 x I _d)		
	- Impulse Withstand (@ ambient temp)	Table 2 (9) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))		
	- AC Withstand	Table 2 (10) (15min x 2.5U ₀)		
	- Humidity	Table 2 (12) (300hrs x 1.25U ₀)		
	- Examination	Table 2 (13)		
	- Range of Compliance	Clause 6 Tables 8 & 9		

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Specification for 11kV and 20kV Cable Joints and Terminations - I/D XLPE Terminations				
	Clause / Requirements	Conformance Code	Remarks / Comments	
Connectors / Lugs	C79			
I/D XLPE Terminations	(HD 629.1)			
- DC Withstand	Table 3 (1) (15min x 6U ₀)			
- AC Withstand	Table 3 (2) (5min x 4.5U ₀)			
- PD (@ ambient)	Table 3 (3) (<10pC @ 2U ₀)			
- Impulse Withstand (@ elevated temp)	Table 3 (4) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- Heat Cycle at Voltage	Table 3 (5) (126 cycles x 2.5U ₀)			
- PD (ambient & elevated temp)	Table 3 (6) (<10pC @ 2U ₀)			
- Thermal S/C-screen	Table 3 (7) (2 x I _{sc})			
- Thermal S/C-conductor	Table 3 (8) (2 x Θ_{sc})			
- Dynamic S/C (Class 1)	Table 3 (9) (1 x I _d)			
- Impulse Withstand (@ ambient temp)	Table 3 (10) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- AC Withstand	Table 3 (11) (15min x 2.5U ₀)			
- Humidity	Table 3 (12) (300hrs x 1.5U ₀)			
- Examination	Table 3 (13)			
- Range of Compliance	Clauses 6.1, 6.4 & 6.6 Tables 9 & 10			

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Specification for 11kV and 20kV Cable Joints and Terminations - Separable Connectors				
	Clause / Requirements	Conformance Code	Remarks / Comments	
Connectors / Lugs	C79			
Separable Connectors				
- Bushing Profile	BSEN 50180			
- Bushing Profile	BSEN 50181			
Separable Connectors	(HD 629.1)			
- DC Withstand	Table 7 (1) (15min x 6U ₀)			
- AC Withstand	Table 7 (2) (5min x 4.5U ₀)			
- PD (@ ambient)	Table 7 (3) (<10pC @ 2U ₀)			
- Impulse Withstand (@ elevated temp)	Table 7 (4) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- Heat Cycle at Voltage	Table 7 (5) (63 cycles x 2.5U ₀ -in air)			
- Heat Cycle at Voltage	Table 7(6) (63 cycles x 2.5U ₀ -in water)			
- Thermal S/C-screen	Table 7 (7) (2 x I _{sc})			
- Thermal S/C-conductor	Table 7 (8) (2 x Θ_{sc})			
- Dynamic S/C (Class 1)	Table 7 (9) (1 x I _d)			
- Connect/ Disconnect	Table 7 (10) (5 complete operations)			
- PD (ambient & elevated temp)	Table 7 (11) (<10pC @ 2U ₀)			

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Specification for 11kV and 20kV Cable Joints and Terminations - Separable Connectors				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Separable Connectors	(HD 629.1)		
-	Impulse Withstand (@ ambient temp)	Table 7 (12) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))		
-	AC Withstand	Table 7 (13) (15min x 2.5U ₀)		
-	Operating Eye	Table 7 (14) (Axial force 1300N / 1min)		
-	Impulse Withstand (@ ambient temp)	Table 7 (15) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))		
-	Examination	Table 7 (16)		
-	Screen Resistance	Table 7 (17) (max 5kΩ)		
-	Leakage Current	Table 7 (18) (max 0.5mA @ U _m)		
-	Screen Fault I Initiation	Table 7 (19) (continuous fault current)		
-	Operating Force	Table 7 (20) (< 900 N)		
-	Capacitive Test Point	Table 7 (21) (test point to cable conductor >1.0 pF, ratio of test point / earth & test point / cable conductor ≤12		
-	Range of Compliance	Clauses 6.1, 6.4 & 6.6 Tables 9 & 10		

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Specification for 11kV and 20kV Cable Joints and Terminations - O/D XLPE Terminations				
	Clause / Requirements	Conformance Code	Remarks / Comments	
Connectors / Lugs	C79			
O/D XLPE Terminations	(HD 629.1)			
- DC Withstand	Table 4 (1) (15min x $6U_0$)			
- AC Withstand	Table 4 (2) (5min x $4.5U_0$)			
- AC Withstand (Wet)	Table 4 (3) (1min x $4U_0$)			
- PD (@ ambient)	Table 4 (4) ($<10pC$ @ $2U_0$)			
- Impulse Withstand (@ elevated temp)	Table 4 (5) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- Heat Cycle at Voltage	Table 4 (6) (126 cycles x $2.5U_0$)			
- Immersion	Table 4 (7) (10 cycles)			
- PD (ambient & elevated temp)	Table 4 (8) ($<10pC$ @ $2U_0$)			
- Thermal S/C-screen	Table 4 (9) ($2 \times I_{sc}$)			
- Thermal S/C-conductor	Table 4 (10) ($2 \times \Theta_{sc}$)			
- Dynamic S/C (Class 1)	Table 4 (11) ($1 \times I_d$)			
- Impulse Withstand (@ ambient temp)	Table 4 (12) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- AC Withstand	Table 4 (13) (15min x $2.5U_0$)			
- Salt Fog	Table 4 (14) (1000hrs x $1.25U_0$)			
- Examination	Table 4 (15)			
- Range of Compliance	Clauses 6.1, 6.4 & 6.6 Tables 9 & 10			

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Specification for 11kV and 20kV Cable Joints and Terminations - Paper & Transition Joints				
	Clause / Requirements	Conformance Code	Remarks / Comments	
Connectors / Lugs	C79			
Sheath (Earth) Connectors	C93			
Paper & Transition Joints	(HD 629.2)			
- DC Withstand	Table 4 (1) (15min x 6U ₀)			
- AC Withstand	Table 4 (2) (5min x 4.5U ₀)			
- Impact	Table 4 (3) (IR: Conductor/Screen ≥10MΩ, Screen/Water ≥50MΩ)			
- Impulse Withstand (@ elevated temp)	Table 4 (4) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- Heat Cycle at Voltage	Table 4 (5) (63 cycles x 1.5U ₀ -in air)			
- Heat Cycle at Voltage	Table 4 (6) (63 cycles x 1.5U ₀ -in water)			
- AC Withstand	Table 4 (7) (4hrs x 3U ₀)			
- Thermal S/C-screen	Table 4 (8) (2 x I _{sc})			
- Thermal S/C-conductor	Table 4 (9) (2 x Θ _{sc})			
- Dynamic S/C (Class 1)	Table 4 (10) (1 x I _d)			
- Impulse Withstand (@ ambient temp)	Table 4 (11) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- AC Withstand	Table 4 (12) (15min x 2.5U ₀)			
- Examination	Table 4 (13)			
- Range of Compliance	Clause 6 Tables 8 & 9			
- Water Withstand	Clause 5.6 (10kPa / 0.1bar)			

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Specification for 11kV and 20kV Cable Joints and Terminations - XLPE Joints				
	Clause / Requirements	Conformance Code	Remarks / Comments	
Connectors / Lugs	C79			
Sheath (Earth) Connectors	C93			
XLPE Joints	(HD 629.1)			
- DC Withstand	Table 5 (1) (15min x 6U ₀)			
- AC Withstand	Table 5 (2) (5min x 4.5U ₀)			
- PD (@ ambient)	Table 5 (3) (<10pC @ 2U ₀)			
- Impact	Table 5 (4) (IR: Conductor/Screen ≥10MΩ, Screen/Water ≥50MΩ)			
- Impulse Withstand (@ elevated temp)	Table 5 (5) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- Heat Cycle at Voltage	Table 5 (6) (63 cycles x 2.5U ₀ -in air)			
- Heat Cycle at Voltage	Table 5 (7) (63 cycles x 2.5U ₀ -in water)			
- PD (ambient & elevated temp)	Table 5 (8) (<10pC @ 2U ₀)			
- Thermal S/C-screen	Table 5 (9) (2 x I _{sc})			
- Thermal S/C-conductor	Table 5 (10) (2 x Θ _{sc})			
- Dynamic S/C (Class 1)	Table 5 (11) (1 x I _d)			
- Impulse Withstand (@ ambient temp)	Table 5 (12) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- AC Withstand	Table 5 (13) (15min x 2.5U ₀)			
- Examination	Table 5 (14)			
- Range of Compliance	Clauses 6.1, 6.4 & 6.6 Tables 9 & 10			
- Water Withstand	Clause 5.6 (10kPa / 0.1bar)			

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Specification for 11kV and 20kV Cable Joints and Terminations - Paper Stop Ends				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Sheath (Earth) Connectors	C93		
	Paper Ends	(HD 629.2)		
	- DC Withstand	Table 5 (1) (15min x 6U ₀)		
	- AC Withstand	Table 5 (2) (5min x 4.5U ₀)		
	- Impulse Withstand	Table 5 (3) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))		
	- AC Voltage (air)	Table 5 (4) (500hrs x 1.5U ₀)		
	- AC Voltage (water)	Table 5 (5) (500hrs x 1.5U ₀)		
	- Impulse Withstand	Table 5 (6) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))		
	- AC Withstand	Table 5 (7) (4hrs x 3U ₀)		
	- Examination	Table 5 (8)		
	- Range of Compliance	Clause 6 Tables 8 & 9		
	- Water Withstand	Clause 5.6 (10kPa / 0.1bar)		

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Specification for 11kV and 20kV Cable Joints and Terminations - XLPE Stop Ends				
	Clause / Requirements	Conformance Code	Remarks / Comments	
Sheath (Earth) Connectors	C93			
XLPE Stop Ends	(HD 629.1)			
- DC Withstand	Table 6 (1) (15min x 6U ₀)			
- AC Withstand	Table 6 (2) (5min x 4.5U ₀)			
- PD (@ ambient)	Table 6 (3) (<10pC @ 2U ₀)			
- Impulse Withstand	Table 6 (4) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- AC Voltage (air)	Table 6 (5) (500hrs x 2.5U ₀)			
- AC Voltage (water)	Table 6 (6) (500hrs x 2.5U ₀)			
- PD (@ ambient)	Table 6 (7) (<10pC @ 2U ₀)			
- Impulse Withstand	Table 6 (8) (10 impulses of each Polarity – 95kV (11kV) & 125kV (20kV))			
- AC Withstand	Table 6 (9) (15min x 2.5U ₀)			
- Examination	Table 6 (10)			
- Range of Compliance	Clauses 6.1, 6.4 & 6.6 Tables 9 & 10			
- Water Withstand	Clause 5.6 (10kPa / 0.1bar)			